

AI-powered visual inspection of roof and cladding conditions to prioritize repair and refurbishment works.

Construction × Automated quality inspection × Computer vision · OS039 v2 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
53 is the world moving	25 can we play, can we win	MEDIUM 0 named in the evidence	€71.9m observed evidence · per year	NOW budgeted procurement within 1...	L2 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 1.0 published reference(s) in Construction, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

We offer AI-powered visual inspection of roof and cladding conditions to prioritize repair and refurbishment works, targeting facility managers and estates teams in healthcare, defense, and other sectors with large building portfolios. This opportunity exists now because recent tenders for roof repair and asbestos cladding replacement show a clear demand for efficient assessment, while advances in computer vision and edge computing make automated defect detection practical and cost-effective.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€173m €12.8m – €691m	€71.9m €5.3m – €287m	€180k €13k – €717k	observed
Observed: tendered contract value, annualised	€123m €123m – €471m	€879k €879k – €3.3m	€2k €2k – €8k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Construction (DE)	66,084 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Computer vision	3.3 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TIR	2025
Annual value of one engagement	€952k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	5,483 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

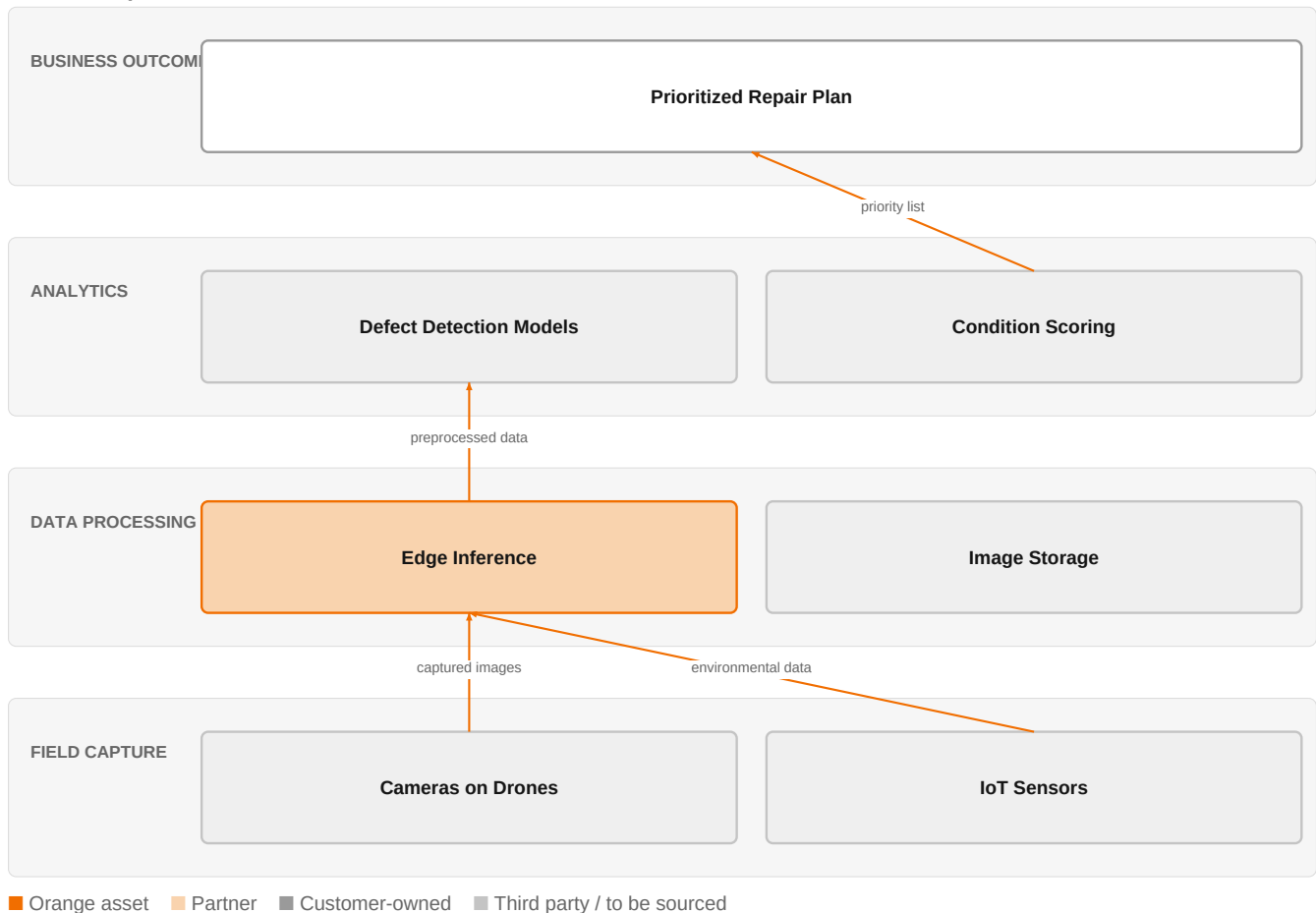
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€123m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	0.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 24 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

AI Roof Inspection Solution



This diagram shows how Orange's IoT and computer vision expertise, combined with NVIDIA edge computing, turns roof imagery into a prioritized repair plan.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent public tenders for roof repair and asbestos cladding replacement indicate a demand for efficient assessment and execution of such works. Advances in computer vision and machine learning now enable real-time defect detection in construction and manufacturing, as demonstrated by studies on earthen additive manufacturing and concrete 3D printing. Edge computing and lightweight algorithms like YOLOv11 make it feasible to deploy these systems on resource-constrained devices, reducing latency and enabling on-site analysis.

Sources: SIG-64EEBA71A3EE contractsfinder.servic 2026-08-17; SIG-DAFB9DCE245C contractsfinder.servic 2026-08-17; SIG-A598AE09F54F openalex.org 2025-11-20; SIG-AE9A7F9C36FF openalex.org 2026-05-04; SIG-5B350B1A3B15 Journal of Organization 2026-05-21

Who buys, and why now

The buyer is typically a facilities director or estates manager responsible for maintaining large building portfolios, often in healthcare, defense, or public sector organizations. The pain is the time and cost of manual roof inspections, which are slow, subjective, and often require scaffolding or drones. The trigger is a budgeted procurement for roof repair or cladding replacement, where they need to prioritize which sections to address first based on condition data.

Why it is hot now — every claim bound to a dated source

- Vision-based quality monitoring with annotated datasets enables deep learning models to detect defects in real-time during earthen 3D printing. [SIG-A598AE09F54F]
- Real-time vision-based defect detection is enabled for large-scale earthen additive manufacturing with annotated datasets. [SIG-A598AE09F54F]

- 3D printing of concrete structures presents challenges due to defects, and AI-driven techniques are reviewed to identify surface defects. [SIG-AE9A7F9C36FF]
- A dual-model framework enables real-time vision-based defect detection for large-scale earthen additive manufacturing with an annotated dataset. [SIG-A598AE09F54F]
- Recent tenders for roof repair and asbestos cladding replacement indicate a demand for efficient assessment and execution of such works. [SIG-64EEBA71A3EE, SIG-DAFB9DCE245C]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution combining its IoT experts and Orange Group researchers to deploy computer vision technology on edge devices (e.g., NVIDIA-powered cameras) that capture roof and cladding imagery. The system would use annotated datasets to train defect detection models, enabling real-time identification of cracks, corrosion, or asbestos deterioration. The engagement would start with a pilot on a single building, then scale to portfolio-wide assessments, integrating with the customer's maintenance management system.

Why Orange can win

Orange can win by leveraging its IoT expertise and research capabilities to deliver a tailored, end-to-end solution that combines edge computing with computer vision. The partnership with NVIDIA provides access to advanced GPU technology for on-device inference. However, the assets are thin: there is no specific Orange-built computer vision platform, so the solution would rely heavily on partner technology and integration skills.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Computer vision	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.7; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Bosch	Industrial / OT specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
Dedrone by Axon	Security specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
HENSOLDT	Security specialist	sells Computer vision	structural	—
Honeywell	Industrial / OT specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	active in Construction; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	active in Construction; competes in OX: Smart Industries	structural	—
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	structural	—
Accenture	Global systems integrator	competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently assess roof and cladding conditions across your portfolio, and what is the typical cycle time?

- 2 Do you have a budgeted procurement for roof repair or cladding replacement in the next 12 months?
- 3 What is the size and age of your building portfolio, and are there known issues with asbestos or flat roofs?
- 4 Do you have existing IoT or sensor infrastructure on your sites that could support edge computing?
- 5 Who is responsible for prioritizing repair works, and what data do they rely on today?
- 6 Are you open to a pilot on a single building to validate the technology before scaling?

Objections you will meet

They will say	You can answer
We already use drones or manual inspections; why do we need AI?	Manual inspections are time-consuming and subjective, and drones still require human review. AI can analyze imagery in real-time, providing consistent, objective defect detection and prioritization, reducing the time to actionable insights.
How accurate is the AI in detecting defects on different roof types?	The accuracy depends on the training data. We would start with a pilot on your specific roof types to fine-tune the models, and we can leverage annotated datasets from similar construction materials to improve performance.
We already have contracts with inspection companies; this would disrupt that.	We can complement existing contracts by providing data that helps you better scope and prioritize the work, potentially reducing the number of site visits and enabling more competitive tendering for repair works.
What about the cost of deploying edge devices across our portfolio?	We can start with a pilot on a single building to demonstrate value, then scale based on ROI. Edge devices are becoming more cost-effective, and the savings from prioritized repairs can offset the investment.

Risks and unknowns

The main risk is that the customer may not see the value of AI inspection over traditional methods, especially if they have existing drone or manual inspection contracts. There is also uncertainty about the accuracy of defect detection models on varied roof materials and conditions, and whether the solution can integrate with legacy maintenance systems. The lack of a proven Orange-specific computer vision offering may make customers hesitant to choose Orange over established industrial players.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief that differentiates our offering by combining Orange's IoT expertise and computer vision capabilities, referencing a partnership with NVIDIA to enhance defect detection accuracy.
Sales	In a meeting with a construction firm's operations director, say: 'We can help you prioritize roof and cladding repairs using AI-powered visual inspection, reducing the need for manual surveys.'
Strategist / Innovator	Commission a deep-dive brief on computer-vision defect detection for construction materials, leveraging Orange Group researchers and IoT experts, and explore a joint proof-of-concept with NVIDIA.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-64EEBA71A3EE	2026-08-17	contractsfinder.servi ce.g...	1	CA18353 - R9321 - SHGH - Roof Works 2
SIG-DAFB9DCE245C	2026-08-17	contractsfinder.servi ce.g...	1	SUPPLY CHAIN NOTICE - : WC1857703 - (EP0060) Replace the asbestos wall and roof cement cladding...
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-AE9A7F9C36FF	2026-05-04	openalex.org	1	Computer Vision and Machine Learning Approaches for Defect Detection in 3D-Printed Cementitious...
SIG-8CA4FFD436BC	2026-03-27	openalex.org	1	Adapting Vision–Language Models for Few-Shot Industrial Defect Detection
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-3E11F4C71251	2025-11-13	openalex.org	1	Artificial Intelligence (AI) in Construction Safety: A Systematic Literature Review
SIG-E9AF6D858220	2025-10-21	openalex.org	1	Review on Application of Machine Vision-Based Intelligent Algorithms in Gear Defect Detection
SIG-1EDDEA256F85	2025-10-16	openalex.org	1	Evaluation of computer vision techniques for quality inspection in casting manufacturing process
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-D46EBCBB2C04	2025-09-16	openalex.org	1	Embedded vision transformers for industrial welding defect detection
SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:42:43+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.